DEPARTMENT OF Computer Science and Engineering
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(An Autonomous Institution)

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Blockchain Based Farm Marketplace for Transparent Agricultural Trade

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1. INTRODUCTION

Farmers lose a large share of profit to intermediaries, and buyers cannot verify product origin. This project is a cross-platform Blockchain Based Farm Marketplace that connects farmers directly with buyers. Payments are held in a smart contract escrow and all transactions are recorded on Ethereum, so no crypto wallet is needed.

2. PROBLEM STATEMENT & OBJECTIVES

Intermediaries such as agents and wholesalers control pricing, so farmers receive only a small share of the final price. Buyers cannot verify the origin of produce, and manual or centralized records can be altered or disputed. A transparent marketplace is needed that connects farmers directly with buyers, secures payments and makes product history verifiable.

Objectives:
- Direct farmer-to-buyer marketplace
- Escrow smart contract (lock, release on delivery, refund on cancel)
- Blockchain-based traceability without extra hardware
- Simple cross-platform app with admin monitoring.

3.METHODOLOGY WITH ARCHITECTURE DIAGRAM

- Frontend: Expo (React Native)
- Backend: Node.js, Express.js, TypeScript, JWT with role-based access
- Blockchain: Solidity contract on Ethereum (Hardhat), Ethers.js relayer
- Storage: MongoDB (data), Cloudinary (images)

Workflow: Register/Login -> Farmer lists produce -> Buyer browses and orders -> Payment locked in escrow -> Buyer confirms delivery -> Funds released to farmer -> Traceability record available on-chain.

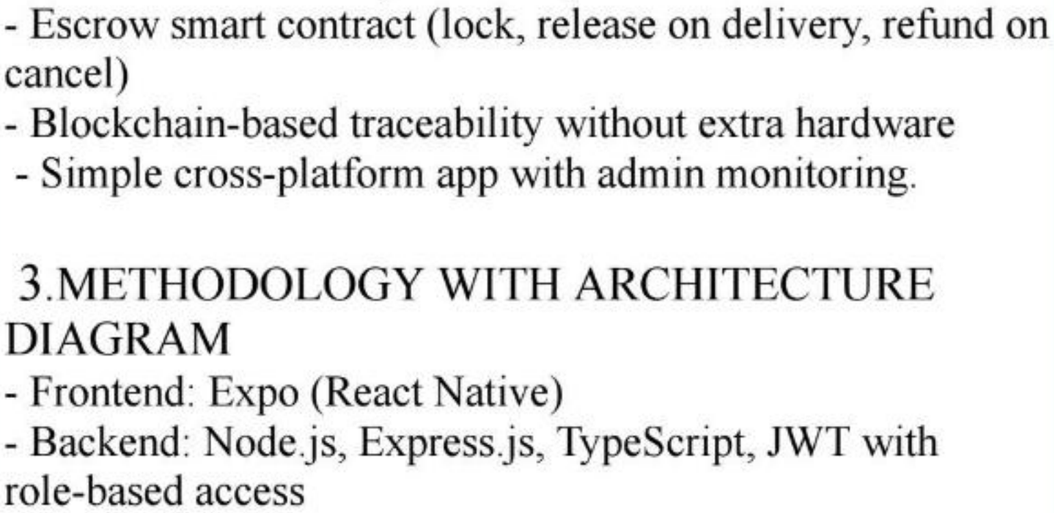
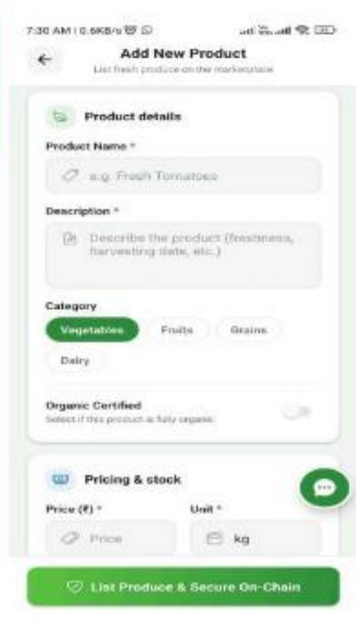


Fig. 2. System architecture of the Farm Marketplace and Agri Agent Logistics System

4. RESULTS / OUTPUT SCREENSHOTS



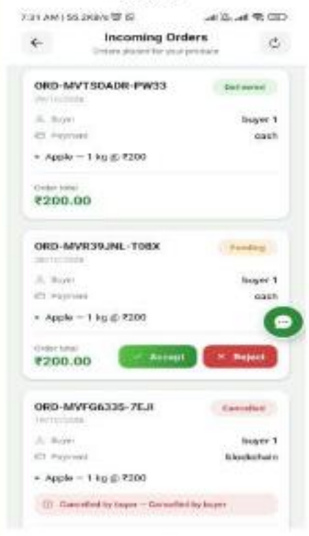
fig(1)



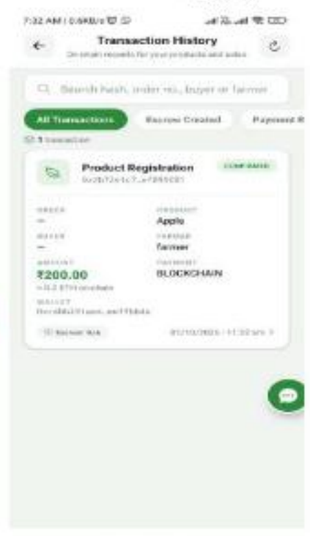
fig(2)



fig(3)



fig(4)



fig(5)

Fig. 1: Register/Login page Fig. 2: Farmer adds a new product Fig. 3: Buyer browses products Fig. 4: Purchase with payment locked in escrow Fig. 5: Transaction recorded on the blockchain

5. CONCLUSION & FUTURE SCOPE

Conclusion:
The system enables direct, escrow-protected, traceable agricultural trade, with the relayer adding some centralization. Future Scope: App store release, IoT monitoring, AI price forecasting, Layer-2 networks, government registry integration, user-controlled wallets.

Future Scope:
- IoT sensor integration for storage and quality monitoring.
- Layer-2 networks (e.g., Polygon) to reduce gas fees.
- Integration with government registries and e-NAM.

6. FUNDING INFORMATION

No external funding. Self-funded student project built on free-tier services

7. POs MAPPED	1, 2, 3, 5, 9, 10, 12
8. SDGs MAPPED	2, 8, 9, 10, 12